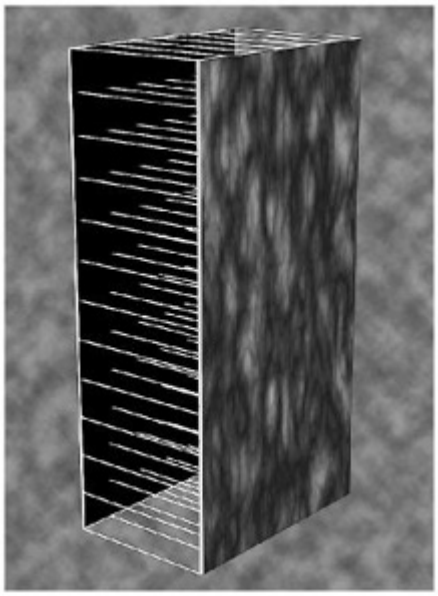


Figure 13.7 Two three-branes, separated by a short interval.

Specifically, Steinhardt and Turok imagine that each brane in Figure 13.7 has three space dimensions, with the line segment between them providing a fourth space dimension. The remaining six space dimensions are curled up into a Calabi-Yau space (not shown in the figure) that has the right shape for string vibrational patterns to account for the known particle species.[10] The universe of which we are directly aware corresponds to one of these three-branes; if you like, you can think of the second three-brane as another universe, whose inhabitants, if any, would also be aware of only three space dimensions, assuming that their experimental technology and expertise did not greatly exceed ours. In this setup, then, another three-brane—another universe—is right next door. It's hovering no more than a fraction of a millimeter away (the separation being in the fourth spatial dimension, as in Figure 13.7), but because our three-brane is so sticky and the gravity we experience so weak, we have no direct evidence of its existence, nor its hypothetical inhabitants any evidence of ours.

But, according to the cyclic cosmological model of Steinhardt and Turok, Figure 13.7 isn't how it's always been or how it will always be. Instead, in their approach, the two three-branes are attracted to each other—almost as though connected by tiny rubber bands—and this implies that each drives the cosmological evolution of the other: the branes engage in an endless cycle of collision, rebound, and collision